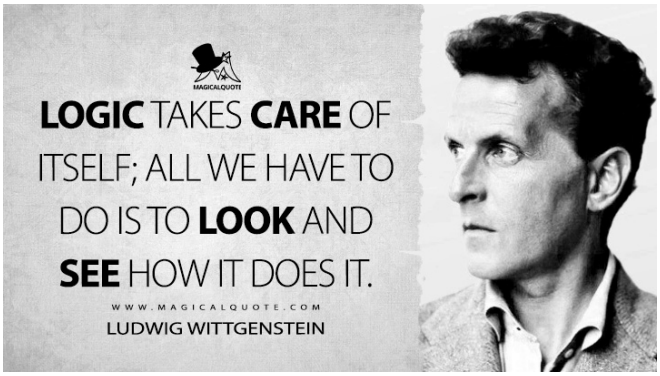
Chapter 1 - Logic and Proof

Note 2

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1.2 : Quantifiers

In note 1, we found that the sentence $x^2 - 5x + 6 = 0$ needed to be considered within a particular context in order to become a statement. When a sentence involves a variable such as x, it is customary to use functional notation when referring to it. Thus we write

$$p(x) : x^2 - 5x + 6 = 0$$

to indicate that $p(x)$ is the sentence "$x^2 - 5x + 6 = 0$". For a specific value of x, $p(x)$ becomes a statement that is either true or false. For example, $p(2)$ is true and $p(4)$ is false.

When a variable is used in an equation or an inequality, we assume that the general context for the variable is the set of real numbers, unless we are told otherwise. Within this context we may remove the ambiguity of $p(x)$ by using a quantifier. The sentence

$$\text{For every } x,\ x^2 - 5x + 6 = 0$$

is a statement since it is false. In symbols we write $\forall\ x,\ p(x)$, where the **universal quantifier** $\forall$ is read, " For every... ", " For all... ", " For each... ", or a similar equivalent phrase. The sentence

$$\text{There exists an } x \text{ such that } x^2 - 5x + 6 = 0.$$

is also a statement, and it is true. In symbols, we write $\exists\ x \ni\ p(x)$, where the **existential quantifier** $\exists$ is read, " There exists... ", " There is at least one... ", or something equivalent.

The symbol $\ni$ is just a shorthand notation for the phrase "such that".

Example 1 : The statement *There exists a number less than 7* can be written $\exists\ x \ni x < 7$ or in the abbreviated form $\exists\ x < 7$, where it is understood that x is to represent a real number. Sometimes the quantifier is not explicitly written down, as in the statement

If x is greater than 1, then x^2 is greater than 1.

The intended meaning is $\forall\ x,\ \text{if } x > 1,\ \text{then } x^2 > 1$.
In general, if a variable is used in the antecedent of an implication without being quantified, then the universal quantifier is assumed to apply.

Example 2 : Rewrite each statement using $\exists$, $\forall$, and $\ni$, as appropriate.

(a). There exists a positive number x such that $x^2 = 5$.

(b). For every positive number M, there is a positive number N such that $N < 1/M$.

(c). If $n \geq N$, then $|f_n(x) - f(x)| \leq 3$ for all x in A.

(d). No positive number x satisfies the equation $f(x) = 5$.

Solution :

(a). $\exists\ x > 0 \ni x^2 = 5$.

(b). $\forall\ M > 0\ \exists\ N > 0 \ni N < 1/M$.

(c). $\forall\ n$, if $n \geq N$, then $\forall\ x$ in A, $|f_n(x) - f(x)| \leq 3$.

(d). The words "no" and "none" are universal quantifiers in a negative sense. In general, the statement "None of them are $P(x)$ " is equivalent to "All of them are not $P(x)$ ". Thus the statement can be written as "$\forall\ x > 0,\ f(x) \neq 5$ ".

Having seen several examples of how existential and universal quantifiers are used, let us now consider how quantified statements are negated. Consider the statement

Everyone in the room is awake.

What condition must apply to the people in the room in order for the statement to be false? Must everyone be asleep? No, it is sufficient that at least one person be asleep. On the other hand, in order for the statement

Someone in the room is asleep.

to be false, it must be the case that everyone is awake. Symbolically, if

$$p(x) : x \text{ is awake,}$$

then

$$\sim [\forall\ x,\ p(x)] \Longleftrightarrow [\exists\ x \ni \sim\ p(x)],$$

where the symbol "$\sim$" represents negation. Similarly,

$$\sim [\exists\ x \ni\ p(x)] \Longleftrightarrow [\forall\ x,\ \sim\ p(x)].$$

Example 3 : Let us look at several more quantified statements and derive their negations. Notice in part (b) that the inequality "$0 < g(y) \leq 1$ " is a conjunction of two inequalities " $0 < g(y)$ " and "$g(y) \leq 1$". Thus its negation is a disjunction. In part (c), note that the "and" between x and y is not a logical connective that needs to be negated. That is, the negation of "for all x and y in A " is "there exist x and y in A". In a complicated statement like (d), it is helpful to work through the negation one step at a time. Fortunately, (d) is about as messy as it will get.

(a). **Statement :** For every x in A, $f(x) > 5$.

$$\forall\ x \text{ in } A,\ f(x) > 5$$

Negation : $\exists\ x$ in $A \ni f(x) \leq 5$. (There is an x in A such that $f(x) \leq 5$.)

(b). **Statement :** There exists a positive number y such that $0 < g(y) \leq 1$.

$$\exists\ y > 0 \ni\ 0 < g(y) \leq 1$$

Negation : $\forall\ y > 0,\ g(y) \leq 0$ or $g(y) > 1$. (For every positive number y, $g(y) \leq 0$ or $g(y) > 1$).

(c). **Statement :** For all x and y in A, there exists z in B such that $x + y = z$.

$$\forall\ x \text{ and } y \text{ in } A, \exists\ z \text{ in } \ B \ni x + y = z.$$

Negation : $\exists\ x$ and y in $A \ni\ \forall\ z$ in $B,\ x + y \neq z$. (There exist x and y in A such that for all z in $B, x + y \neq z$).

(d). **Statement :** $\forall\ \varepsilon > 0\ \exists\ N \ni \forall\ n$, if $n \geq N$, then $\forall\ x$ in $S, |f_n(x) - f(x)| < \varepsilon$

Negation :

$\exists\ \varepsilon > 0 \ni \sim [\exists\ N \ni \forall\ n$, if $n \geq N$, then $\forall\ x$ in $S,\ |f_n(x) - f(x)| < \varepsilon]$,

or

$\exists\ \varepsilon > 0 \ni \forall\ N, \sim [\forall\ n$, if $n \geq N$, then $\forall\ x$ in $S, |f_n(x) - f(x)| < \varepsilon]$,

or

$\exists\ \varepsilon > 0 \ni \forall\ N\ \exists\ n \ni \sim [$ if $n \geq N$, then $\forall\ x$ in $S, |f_n(x) - f(x)| < \varepsilon]$,

or

$\exists\ \varepsilon > 0 \ni\ \forall\ N \exists\ n \ni n \geq N$ and $\sim [\forall\ x$ in $S,\ |f_n(x) - f(x)| < \varepsilon]$,

or

$\exists\ \varepsilon > 0 \ni\ \forall\ N\ \exists\ n \ni n \geq N$ and $\exists\ x$ in $S \ni |f_n(x) - f(x)| \geq \varepsilon$.

Example 4 : Write the negation of each statement in Example 2.

Solution :

(a). $\forall\ x > 0,\ x^2 \neq 5$.

(b). $\exists\ M > 0 \ni \forall\ N > 0,\ N \geq 1/M$.

(c). $\exists\ n \ni n \geq N$ and $\exists\ x$ in $A \ni |f_n(x) - f(x)| > 3$.

(d). $\exists\ x > 0 \ni f(x) = 5$.

It is important to realize that the order in which quantifiers are used affects the truth value. For example, when talking about real numbers, the statement

$$\forall\ x\ \exists\ y \ni y > x$$

is true. That is, given any real number x, there is always a real number y that is greater than x. But the statement

$$\exists\ y \ni \forall\ x, y > x$$

is false, since there is no fixed real number y that is greater than every real number. Thus care must be taken when reading (and writing) quantified statements so that the order of the quantifiers is not inadvertently changed.

Example 5 : Determine the truth value of each statement. Assume that x and y are real numbers. Justify your answers.

(a). $\forall\ x\ \exists\ y \ni x + y = 3$.

(b). $\exists\ x \ni \forall\ y,\ x + y \neq 3$.

Solution :

(a).
- **Statement :** For every x, there exists a y such that $x + y = 3$
- If given any x, can we find a y that makes $x + y$ equal to 3 ? YES !
- "Given any x, let $y = 3 - x$" (This not only states that there is such a y, but it identifies what y is).

(b).
- **Statement :** There exists an x such that for all $y,\ x + y \neq 3$.
- Note that statement (b) is the negation of statement (a).
- Since (a) is true, (b) is false,

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